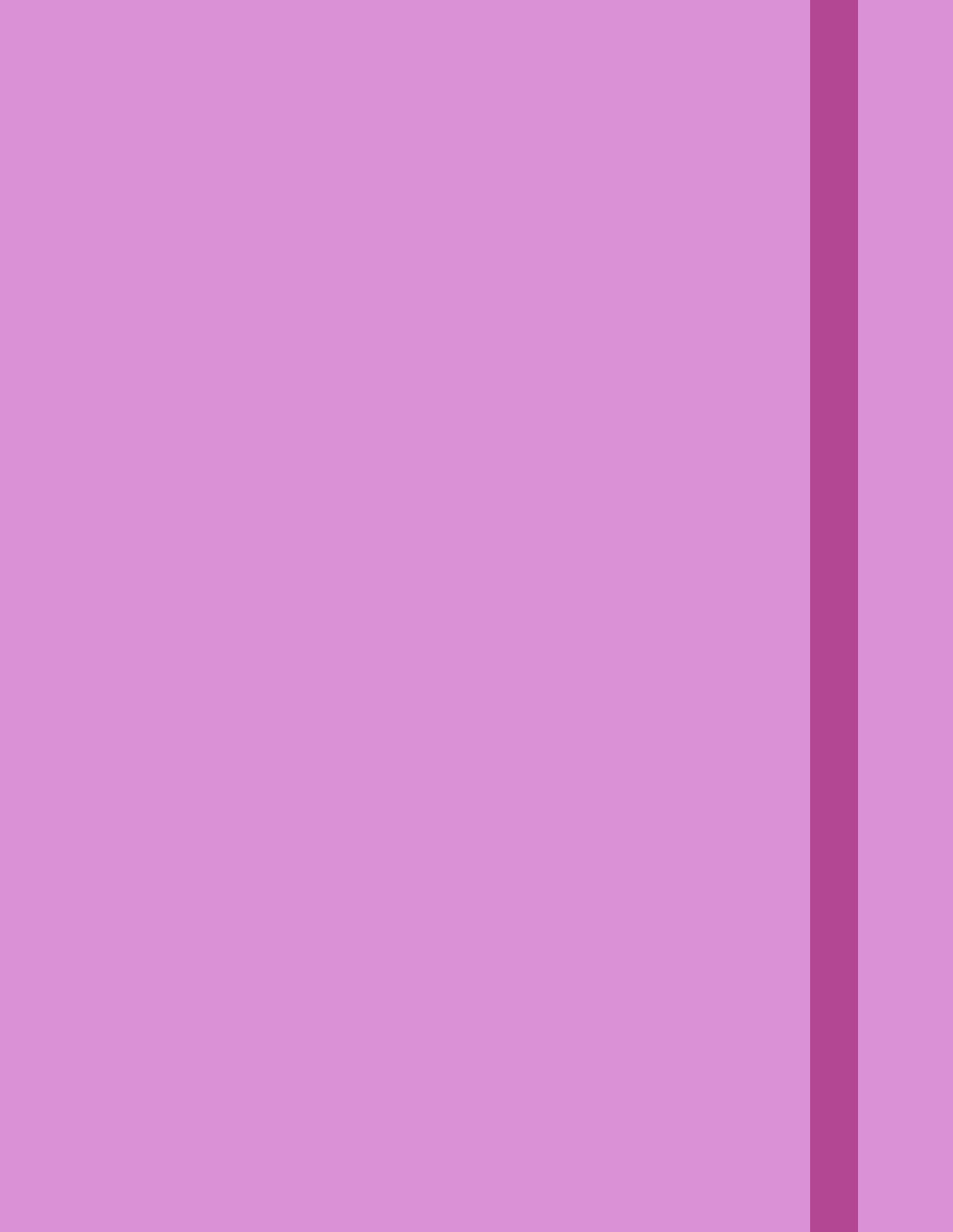






EDA Notebook

0 Imports & Plotting Style

- i don't know what's going on in this first box here but I don't feel like it's really explained

1 Load the Online Dataset

- same here

Understanding the Diamonds Dataset: What You're Researching

- list each feature variable, what it is, why it matters, range/categories, and practical note
- target variable as well, for supervised learning
- need to collect the information you have about the thing to predict how the variables will behave
 - * we're using linear models here, so if something's likely to be
- be careful with outliers *not linear, note it*

2 First Structural Look at the Data

- * inspect:
 1. dimensions
 2. data types
 3. a few rows
 4. missingness
 5. duplicates

what the heck is seaborn?

the data grunt work :)

> I think we're working with a Pandas dataframe here but hard to say when I don't know how seaborn works

```
display(df_work.head())
```

```
df_work.info()
```

```
display(df_work.describe(include="all").T)
```

```
n_duplicates = df_work.duplicated().sum()
```

- make a summary table (also a Pandas dataframe) with number of rows, number of columns, duplicates)

- differentiate between numerical & categorical columns (Zuri Blula would be proud)

```
df_work.select_dtypes(include=np.number).columns.tolist()
```

^{exclude}

I think it might be worth investing the time to see how capitol works

> numeric columns → correlations/regression

> categorical columns → comparisons/encoding

- if a lot of duplicates, need to be careful
- if a lot of missing data, need a missing-data strategy

Univariate Plots of all Variables

- numeric variables $\rightarrow$ histogram & KDE these are just counters, not with respect to anything specific (target variable yaozi)
- categorical variables $\rightarrow$ count plot
- w/ seaborn, easy to do histplot w/ kde

Outlier Finding

- using zero masking still unclear on this
- get rid of points where one dimension of the diamond is zero, since that's probably wrong
- flag if more than 3.5 standard deviations from the mean
- this code is inadequately commented & I'm really not sure what's going on here
- data plots where outliers are in red
- removed if a dimension was equal to zero $\rightarrow$ but not automatic delete with standard deviation rule

Log Transformation

- create log_price & use it as the modeling target for the class confused why sns.pairplot - pairs them all
- then price & logprice histograms
- log so more linear - reduce skewness

Regression on Log Price

- use label encoder on the categorical/object data - idk what this is or whether Zvi Gilula would start screaming at it
- build correlation matrix and drop nan
- correlation heatmap
- $\hookrightarrow$ can I just use their code? All seems like a lot of work

Correlation Analysis: Spearman vs Pearson

- use scipy.stats.spearmanr, pearsonr
- $\hookrightarrow$ compute Pearson & Spearman correlations with price
- $\triangleright$ each have coeff & r

- Pearson measures linear relationship; Spearman measures monotonic relationship $\rightarrow$ look for differences
- the correlation with price is with the whole numeric subset
- these correlations tell you
 - > which variables to use for regression
 - > whether to transform data to make more linear (like taking log)

Regression Model: Prediction & Statistical Inference

One-Hot Encoding

- > for making categorical inputs into numeric (binary) inputs for regression
- > convert each categorical value into binary indicators
- > when n categories $\rightarrow n-1$ dummies $\rightarrow$ no redundancy
 - in other words, one is baseline reference, and the rest are difference from baseline
 - because of intercept, having all of them would make them linearly dependent
- the coefficients indicate log-targetvariable-units from whatever you chose as baseline (= one you drop)
 - > idk why log

`X = pd.get_dummies(X, columns=cats_cats_in_X, drop_first=True, dtype=int)`

Data Quality & Distribution Checks

Missing Values Analysis

- count missing values per column (remember: column-biased database)

`df_work.isnull().sum()`

`df_work.isnull().sum() / len(df_work)`

- check for rows with any missing values

`rows_with_na = df_work.isnull().any(axis=1).sum()`

- strategy for missing data

<5% & random pattern	5-20 %	>20%	Missing data not random
drop rows w/ NaN	imputation (mean for numeric, mode for categorical)	domain expertise or feature engineering	create indicator / flag variable

• Boxplots: Distribution by Categorical Groups

Box: inner quartile range - contains 50% of data

line inside box: 50th percentile

lower box edge: 25th percentile

upper box edge: 75th percentile

→
median line → does price differ by categorical group

box height: variability within each group

whisker length: how spread out

skewness: is the median off-center in the box

outliers

• Ordinary Least Squares

`X = sm.add_constant(X)`

`model = sm.OLS(y, X).fit()`

`print(model.summary())`

↳ look at the residuals

↙
use residuals to remove outliers

check out statsmodels api

↳ allows to easily add constant term column

→ also makes ordinary least squares object: `model = sm.OLS(y, X).fit()` → can use this in seaborn: `sns.residplot(x=model.fittedvalues, y=model.resid)`

→ use seaborn to plot residuals (residuals vs fitted values, and residuals histogram)

→ do something called a qq plot of residuals

→ to remove outliers, use somethings called influence, cook's distance, which are badly explained here → maybe meant to learn about it in regression course?

→ use outlier mask to remove outliers

↳ after cleaning the data, refit the data, and apply replot residuals, qqplot

→ qq plot is theoretical & sample quantiles

Removing Outliers Based on Feature Data

- looking only at continuous numeric columns
- take z scores: $\left| \frac{x - \bar{x}}{\text{std}(x)} \right|$
- keep rows where all z-scores are less than 3 (make a mask using axis=1)
- then refit the model again, `sm.OLS(y_clean_data, X_clean_data).fit()`
replot residuals & qqplot

BUT: there are libraries to take care of all this stuff. sweetviz